杜毅衡 Yiheng Du

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Education

- **Peking University** *Sept 2026 – Jun 2029 (expected)*
  M.S. in *Computer Science and Technology*
  Research Focus: Reinforcement Learning, Agentic RL, Multimodal Learning
- **Sichuan University** *Sept 2022 – Jun 2026*
  B.S. in *Computer Science and Technology*

Experience

Research Intern Mar 2026 – Present
Tencent · Hunyuan · Foundation Model Department

- A main contributor to **UniRL**, a reinforcement learning framework for unified multimodal models. Primarily responsible for the **prompt enhancement (PE) trainer** and **agentic RL** components.
- Developed RL workflows for prompt rewriting with feedback from generated images, and for multi-turn tool-use agents connecting trajectory rollout, reward evaluation, and policy updates.

Research Intern From Mar 2025
Peking University VILLA Lab

- **AlignedGen** (NeurIPS 2025): Equal-contribution first author. Led core method design, pipeline implementation, and main experiments for training-free style-consistent image generation.
- **DDAVS** (ECCV 2026): In collaboration with THU IVG Lab, led core method design, end-to-end implementation, and major experiments for audio-visual segmentation with disentangled audio semantics and delayed bidirectional alignment.

Research Intern Aug 2024 – Feb 2025
Ant Group · Data Intelligence Platform & Services Department

- Contributed to **DLRover** for efficient and reliable distributed training. Designed and implemented **Flash-checkpoint**, and conducted **Auto Accelerate** experiments and analysis for large-scale recommendation model training.

Publications (* denotes equal contribution)

- **AlignedGen: Aligning Style Across Generated Images**, *NeurIPS 2025* (CCF-A)
  Jiexuan Zhang*, **Yiheng Du***, Qian Wang, Weiqi Li, Yu Gu, Jian Zhang.
- **Delayed Bidirectional Alignment via Disentangled Audio Semantics for Audio-Visual Segmentation**, *ECCV 2026*
  Jingqi Tian*, **Yiheng Du***, Haoji Zhang, Yuji Wang, Isaac Ning Lee, Xulong Bai, Tianrui Zhu, Jingxuan Niu, Yansong Tang.
- **DLRover-RM: Resource Optimization for Deep Recommendation Models Training in the Cloud**, *VLDB 2024* (CCF-A)
  Qinlong Wang*, Tingfeng Lan*, Yinghao Tang, Bo Sang, Ziling Huang, **Yiheng Du**, Haitao Zhang, Jian Sha, Hui Lu, Yuanchun Zhou, Ke Zhang, Mingjie Tang.

Awards

- Gold Medal, International Collegiate Programming Contest (ICPC) 2022
- Bronze Medal, International Collegiate Programming Contest East Asia Regional Final (ICPC-EC Final) 2023
- Gold Medal, CCF Collegiate Computer Systems and Programming Contest (CCSP) 2023
- Gold Medal, ICPC China Sichuan Provincial Contest (2nd place in 2024; 3rd place in 2023 & 2022)
- Bronze Medal, National Olympiad in Informatics (NOI) 2021
- Silver Medal, Asia-Pacific Informatics Olympiad (APIO) 2021

Skills

- **Programming:** C/C++, Python, Java
- **Machine Learning:** PyTorch, Transformers, Diffusion Models, Flow Matching, LLMs, RL, Agentic RL
- **Systems:** Linux, Docker, Kubernetes, Distributed Systems, Databases